ARCHITECTURE

Containers Module - Architecture

Version: 2.0.0
Last Updated: February 2026

Design Philosophy

The Containers module provides a **generic, runtime-agnostic** abstraction for container orchestration. It is designed to:

1. **Be reusable** - No dependencies on specific applications
2. **Support multiple runtimes** - Docker, Podman, Kubernetes
3. **Enable remote distribution** - Deploy to multiple hosts via SSH
4. **Be observable** - Health checks, metrics, events
5. **Be safe** - Thread-safe, proper cleanup, graceful shutdown

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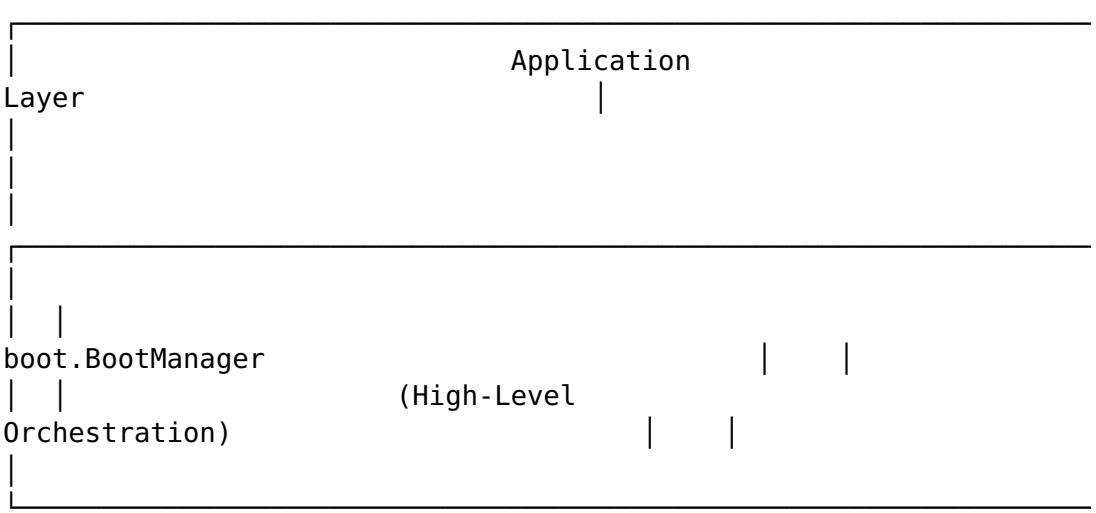
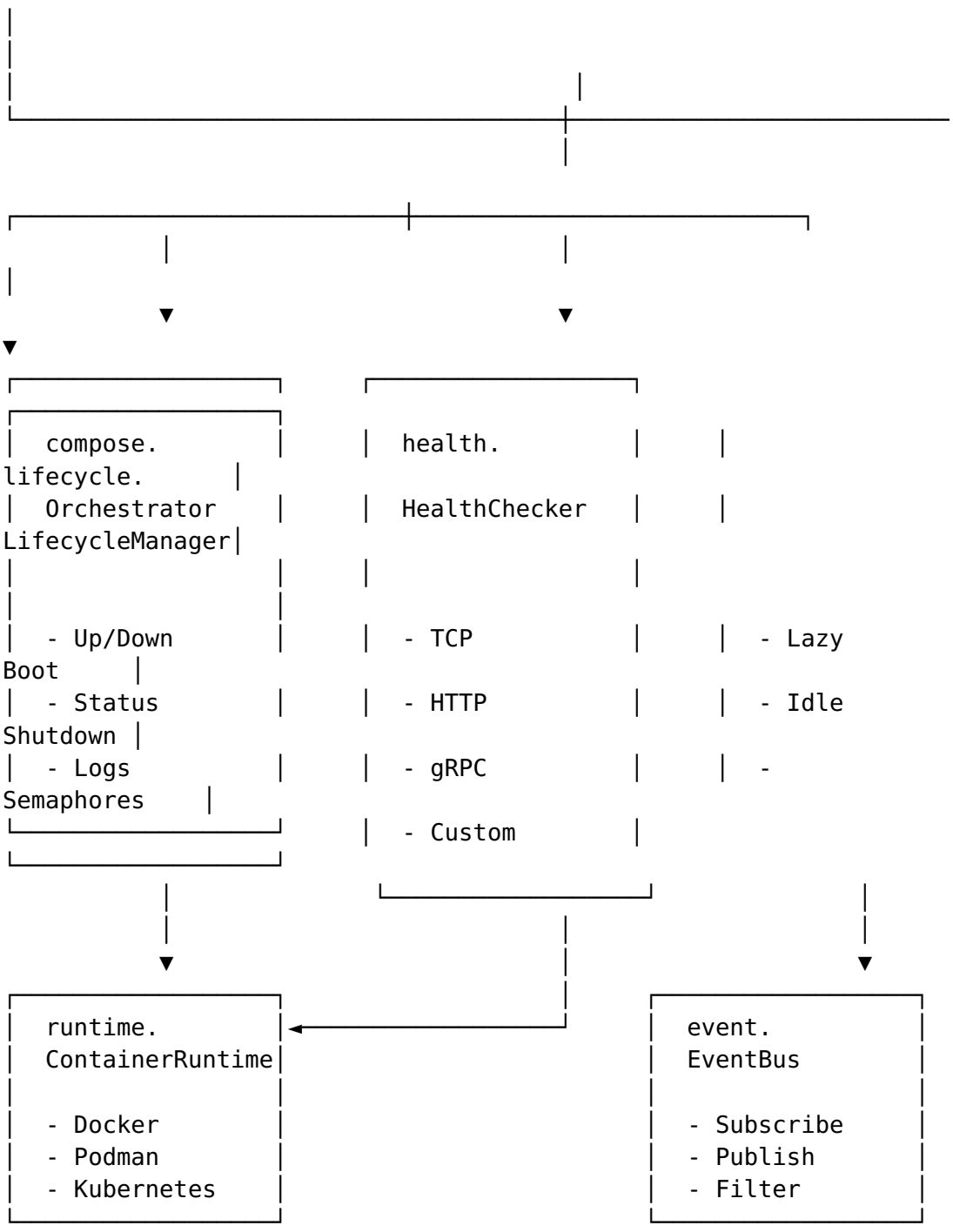
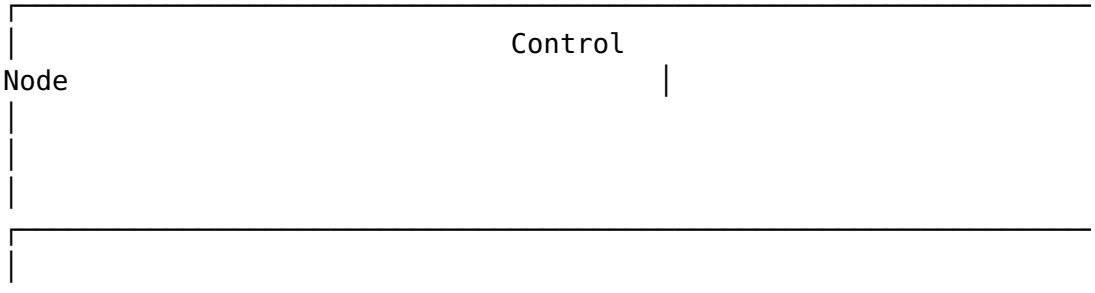
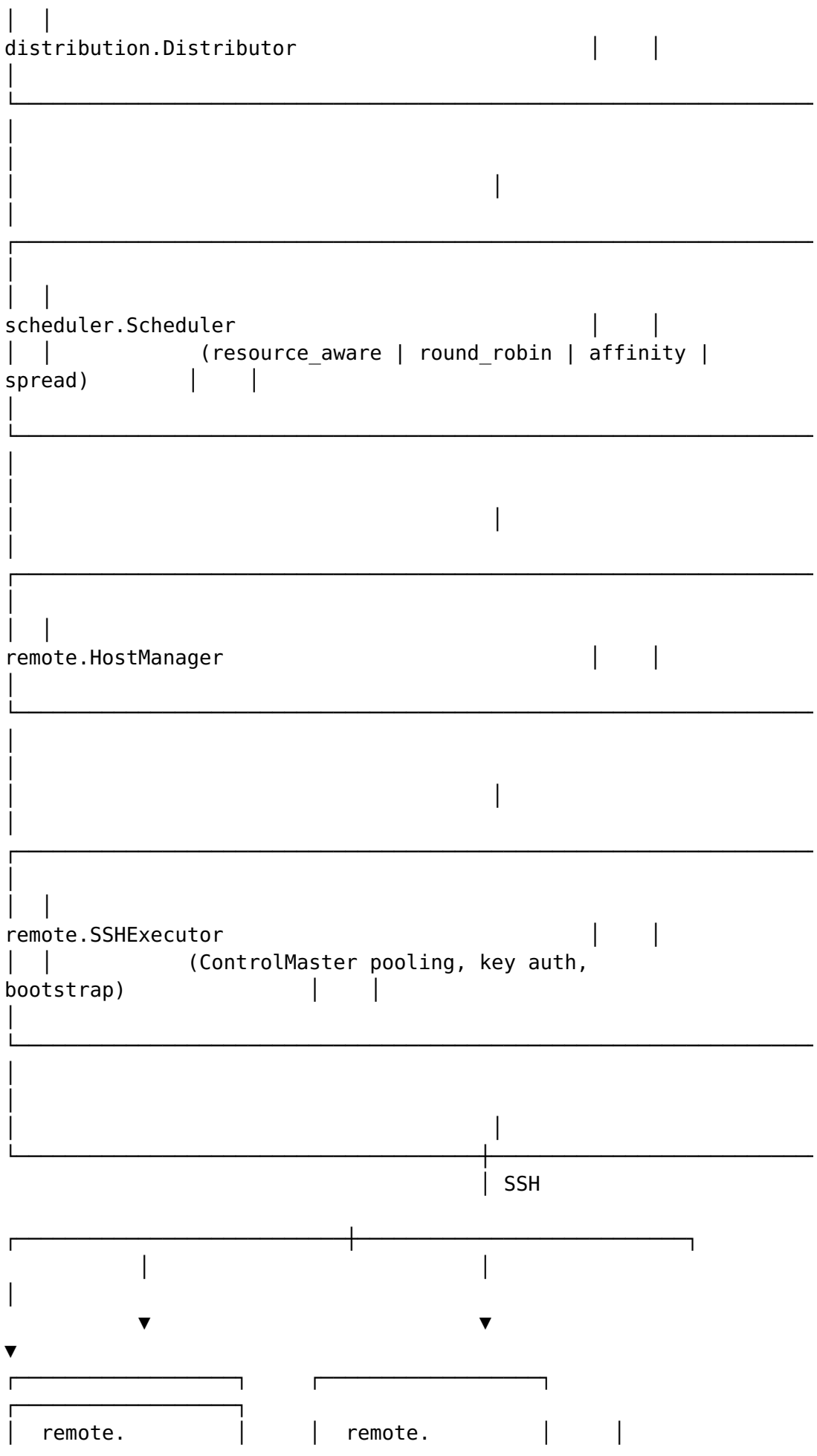
High-Level Architecture

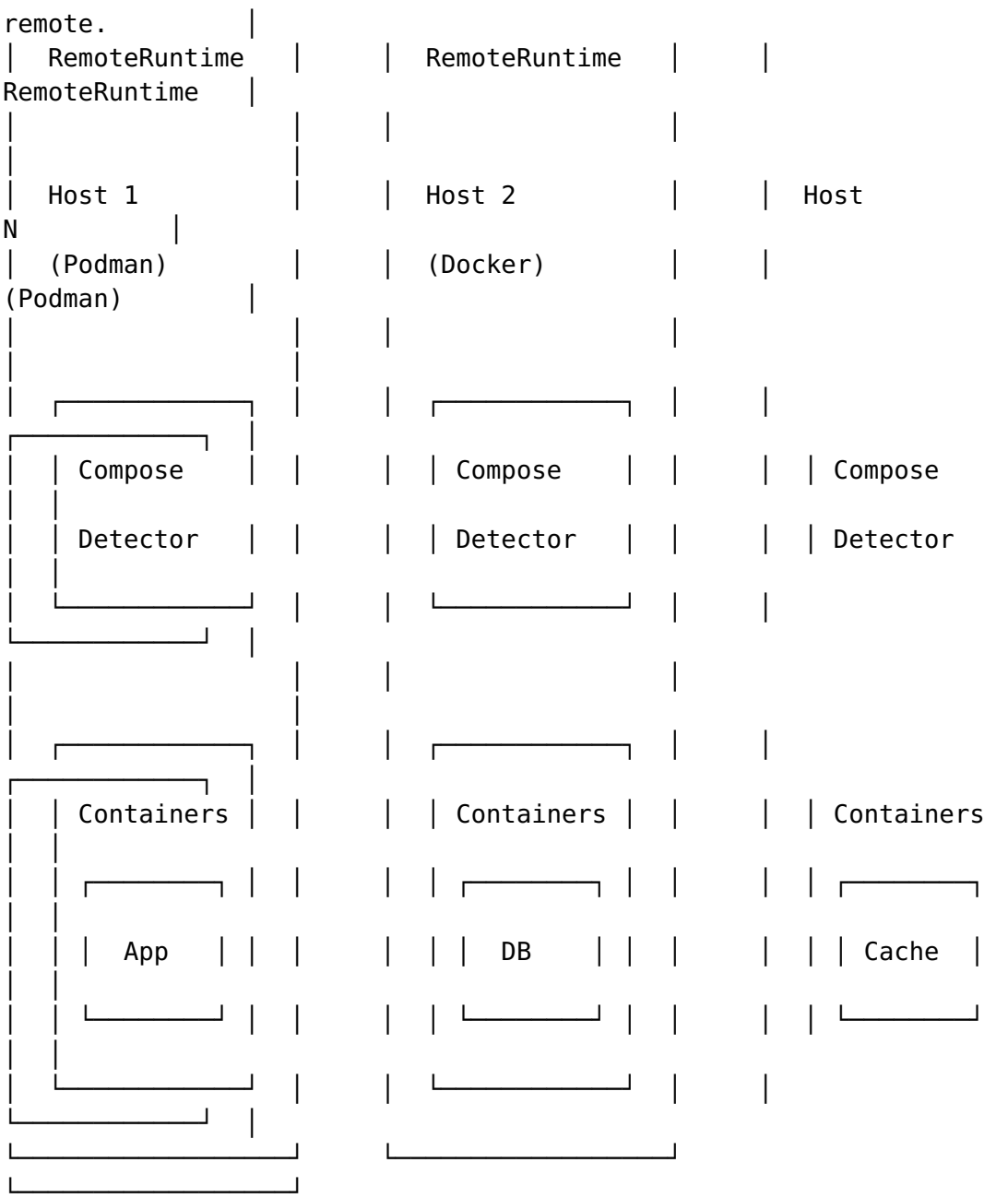




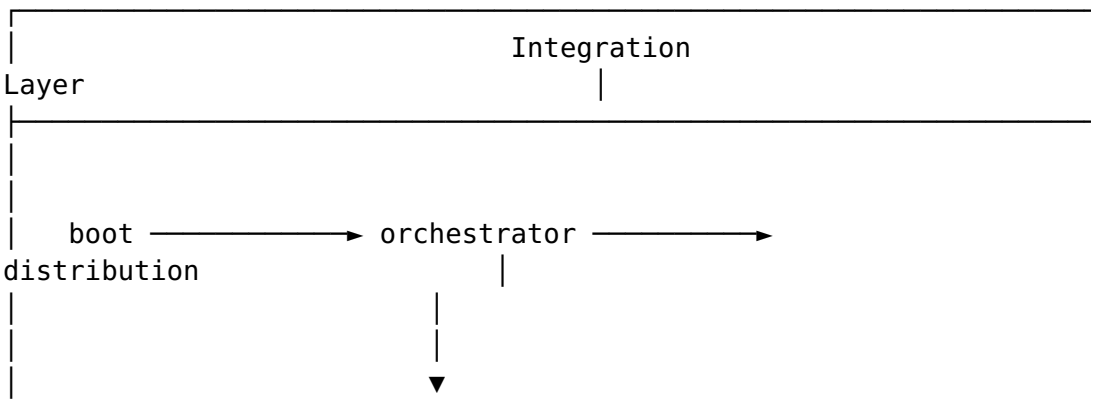
Remote Distribution Architecture

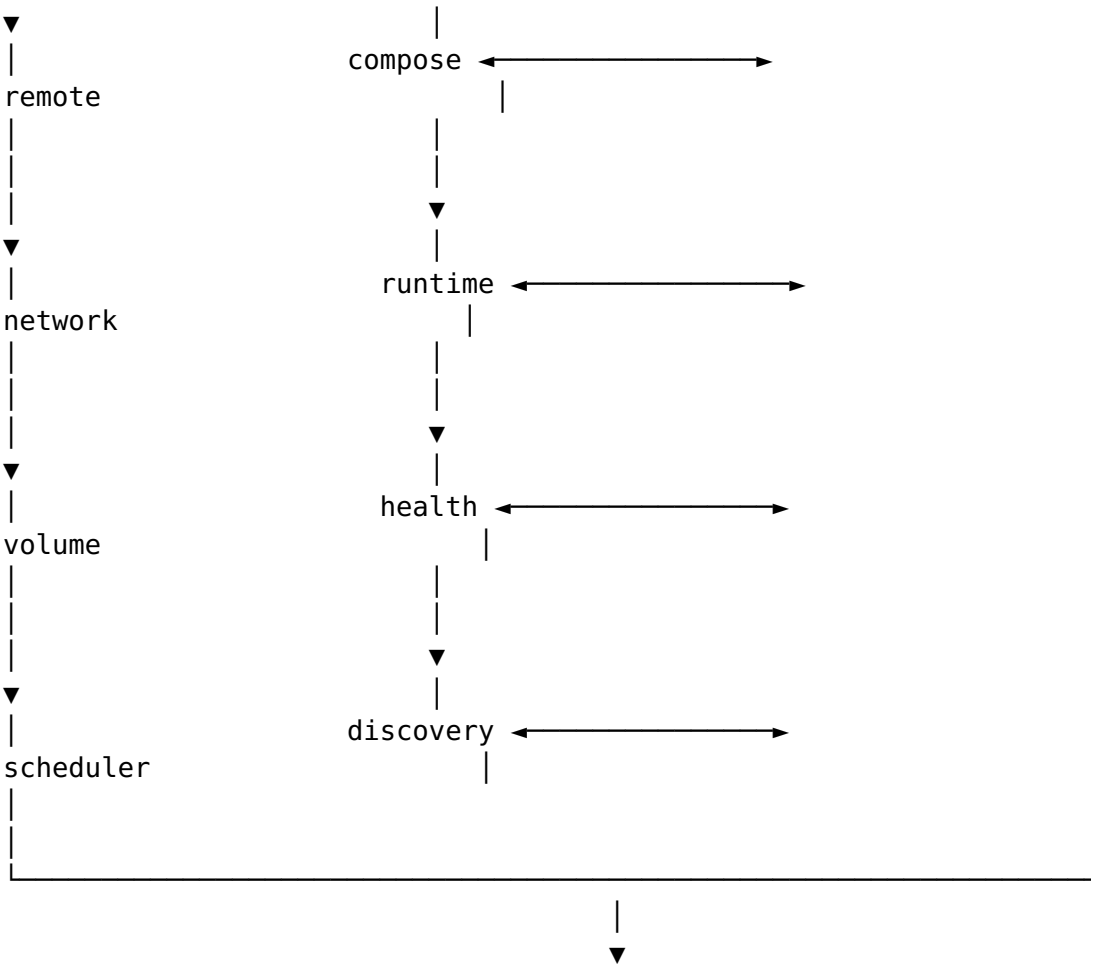


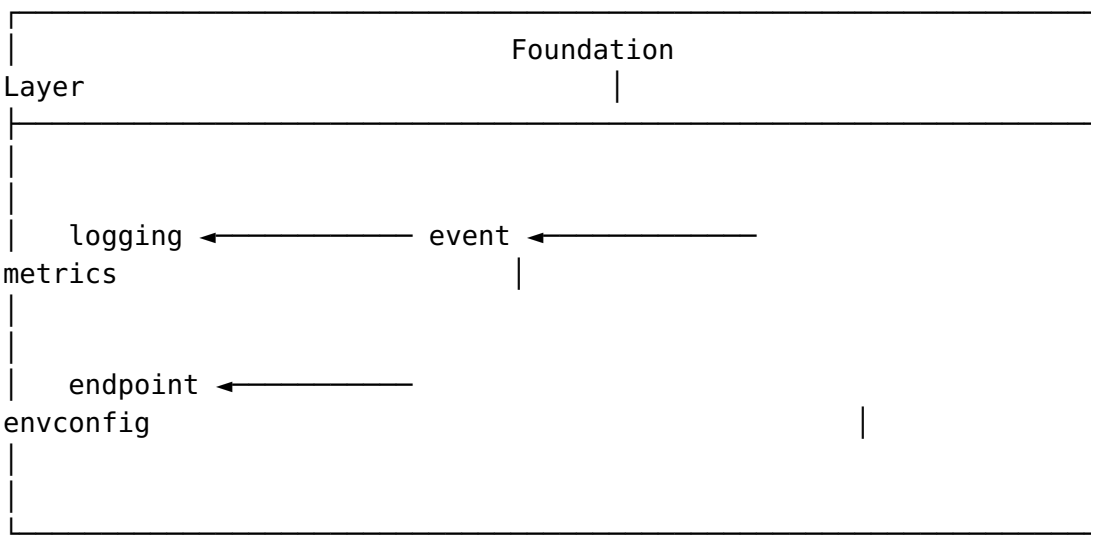

distribution.Distributor
scheduler.Scheduler
(resource_aware | round_robin | affinity | spread)
remote.HostManager
remote.SSHExecutor
(ControlMaster pooling, key auth, bootstrap)
SSH
remote.
remote.



Package Dependency Graph



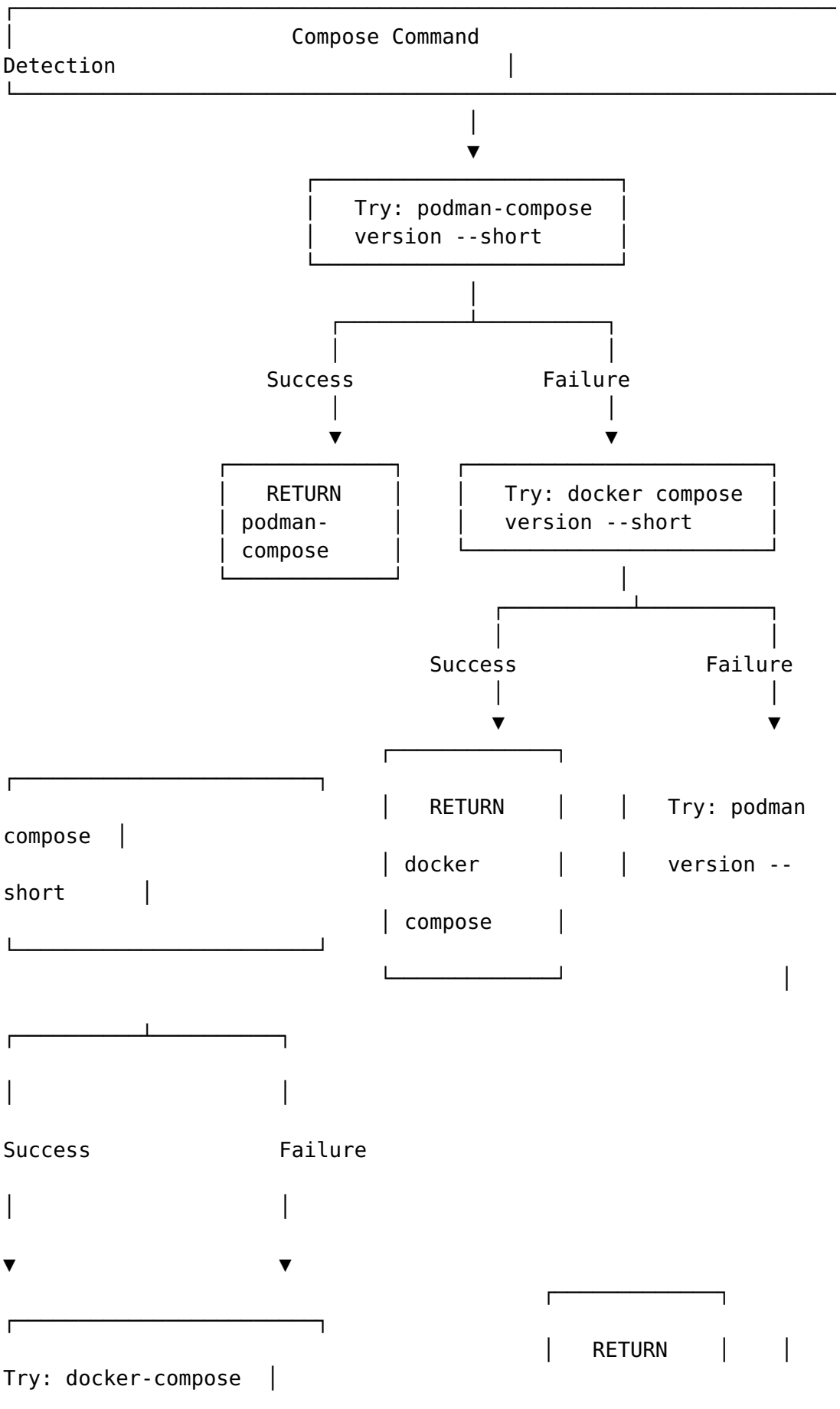


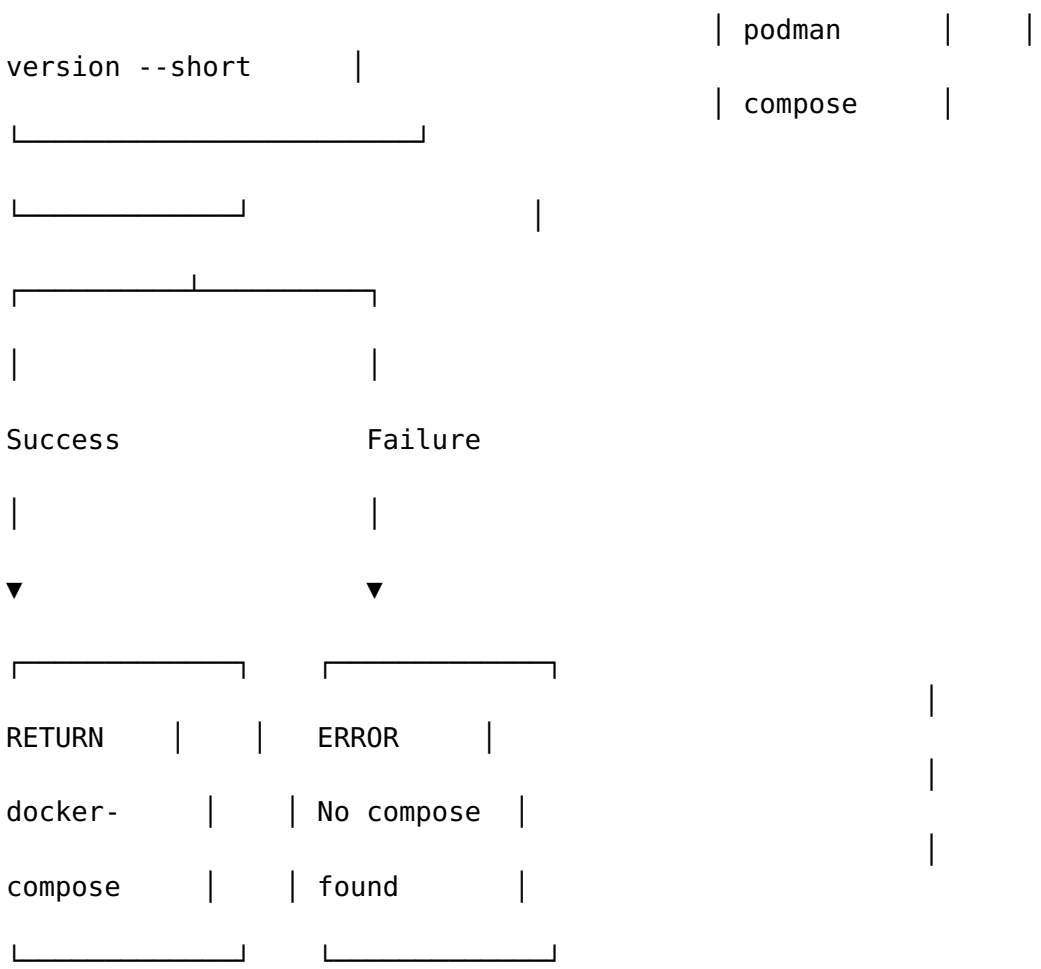


Dependency Rules

1. **No circular dependencies** - The graph is strictly acyclic
2. **Foundation packages have no internal dependencies**
3. **Integration packages may depend on multiple foundation packages**
4. **boot is the top-level integrator**

Compose Detection Flow

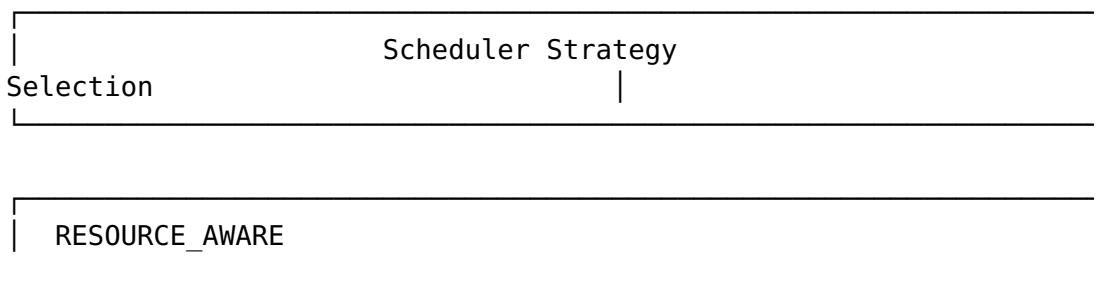




Why podman-compose is Preferred

Command	Issue	Solution
`podman compose`	Delegates to `docker-compose` v1	Use `podman-compose` instead
`docker-compose` v1	Incompatible with Podman (http+docker error)	Install Docker Compose v2 or podman-compose
`podman-compose`	✅ Native Podman support	Preferred for Podman hosts

Scheduler Strategy Flow



```
(Default)                                                    │
│
│
│  1. Probe each host's CPU, Memory, Disk,
Network                              │
│  2. Calculate score: CPU(40%) + Memory(40%) + Disk(10%) +
Network(10%)     │
│  3. Select host with highest available
resources                             │
│
│
│  Use Case: Mixed workloads, optimal resource
utilization                     │
└──────────────────────────────────────────────────────────────────────────────

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│
ROUND_ROBIN
│
│
│
│  1. Maintain index of last used
host                                          │
│  2. Select next host in
rotation                                              │
│  3. Wrap around when reaching
end                                            │
│
│
│  Use Case: Simple load balancing, equal
distribution                          │
└──────────────────────────────────────────────────────────────────────────────

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│
AFFINITY
│
│
│
│  1. Match container labels with host
labels                                    │
│  2. Prefer hosts with matching
labels                                         │
│  3. Fall back to other hosts if no
match                                       │
│
│
│  Use Case: GPU workloads, storage requirements, architecture
matching       │
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│
```
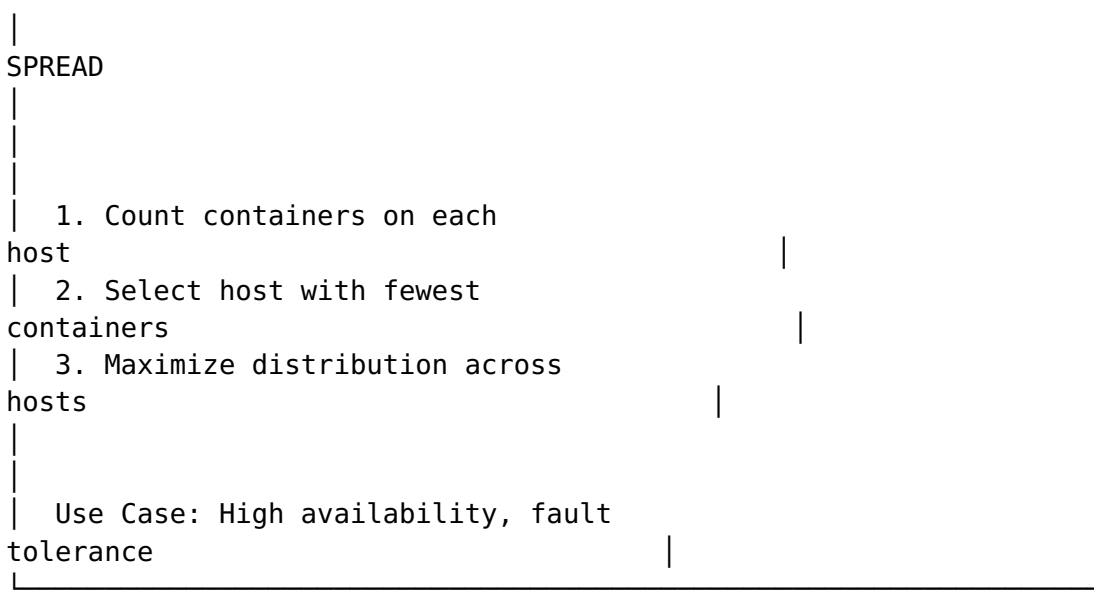
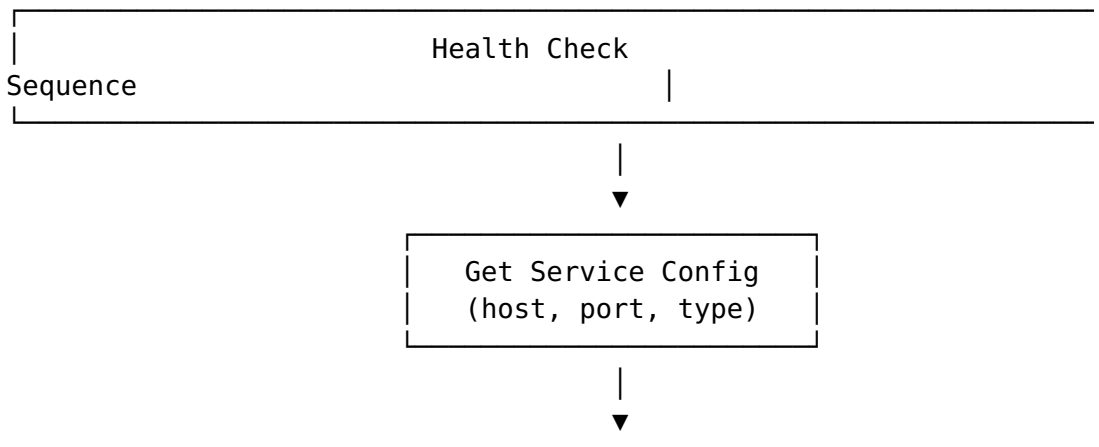
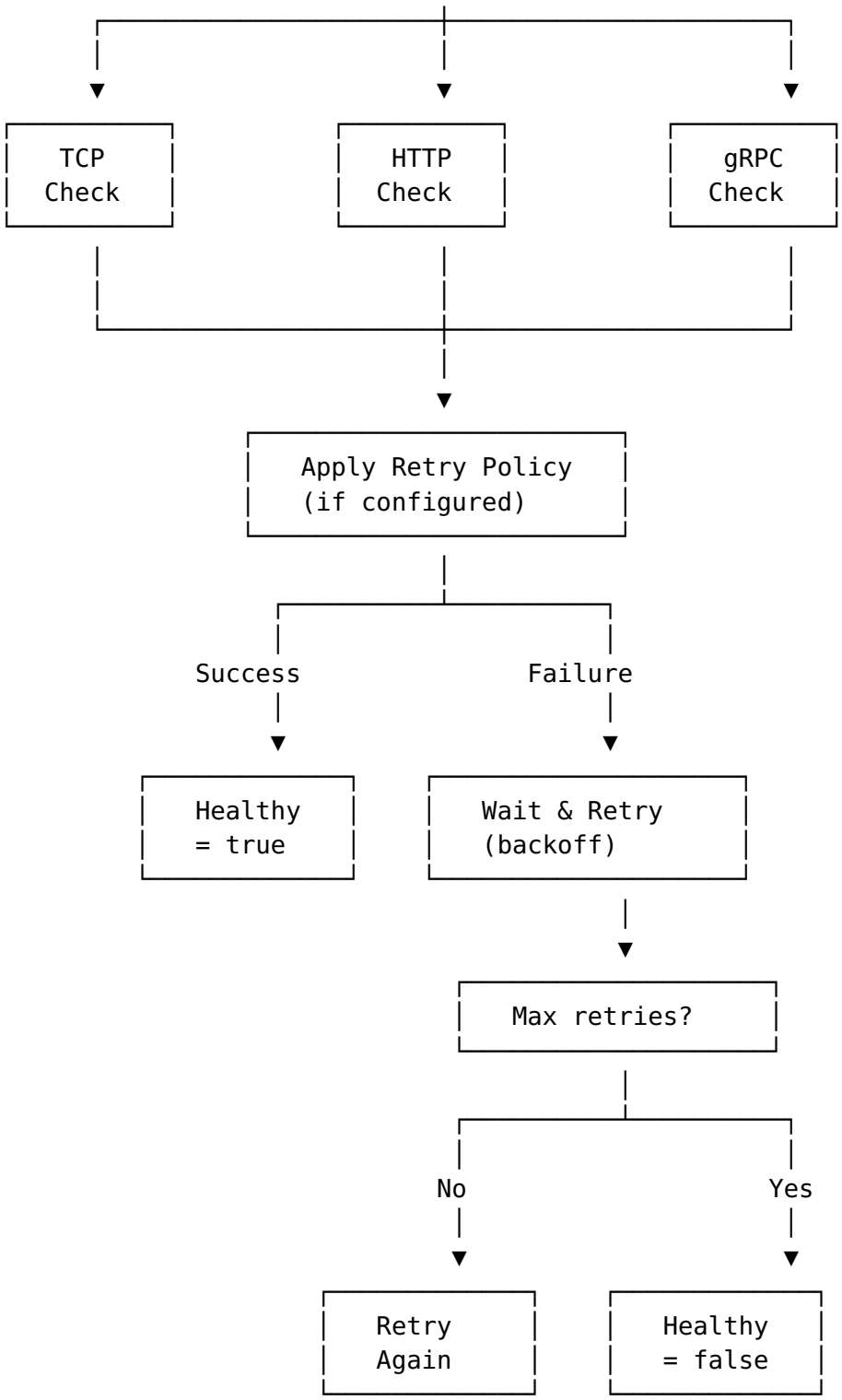
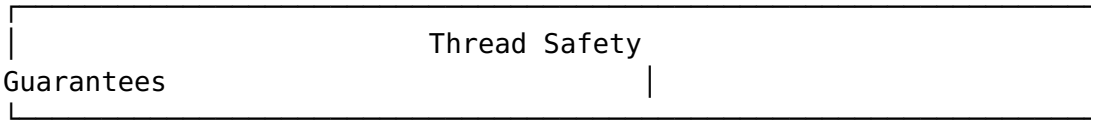



Health Check Flow





Thread Safety Model



Package	Mechanism	Scope
boot.BootManager	sync.RWMutex	Service registry, state
runtime.*Runtime	sync.Mutex per client	API calls
compose.*Orchestr.	Stateless	None needed
health.*Checker	Stateless	None needed
remote.HostManager	sync.RWMutex	Host registry
remote.SSHExecutor	sync.Pool + ControlMstr	Connection pooling
scheduler.*	sync.RWMutex	Host scores
event.*EventBus	Channel-based	Event delivery
monitor.*Monitor	sync.RWMutex	Metrics snapshot
lifecycle.*Manager	sync.Mutex + semaphore	Lifecycle state
distribution.*	sync.RWMutex	Workflow state

Design Patterns Used

Strategy Pattern

- `ContainerRuntime` interface with Docker/Podman/K8s implementations
- `HealthChecker` with TCP/HTTP/gRPC/Custom check strategies
- `Scheduler` with 5 scheduling strategies

Observer Pattern

- `EventBus` publishes lifecycle events (started, stopped, health changed)
- Subscribers filter by event type or source

Factory Pattern

- `runtime.AutoDetect()` creates the appropriate runtime
- `health.NewDefaultChecker()` creates a checker with all strategies

Builder Pattern

- `endpoint.NewEndpoint().WithHost().WithPort().Build()`
- Fluent API for constructing ServiceEndpoint configurations

Decorator Pattern

- `health.RetryPolicy` wraps any health check with retry logic
- Logging wrappers can be applied to any interface

Functional Options

- `boot.NewBootManager(WithRuntime(), WithLogger(), ...)`
- Extensible configuration without breaking changes

Proxy Pattern

- `remote.RemoteRuntime` implements `ContainerRuntime` over SSH
- `remote.RemoteComposeOrchestrator` implements compose over SSH

Boot Sequence

```
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│                              Boot
Sequence                                          │
└──────────────────────────────────────────────────────────────────────────────┘

1. DISCOVERY PHASE
   ├── Probe endpoints with discovery enabled
   └── Mark discovered services (skip compose start)

2. GROUPING PHASE
   ├── Group services by compose file + profile
   └── Separate local vs remote services

3. START PHASE (Local)
   ├── For each compose group:
   │   ├── Execute: docker compose -f <file> --profile <p> up -d
   │   └── Wait for command completion
   └── Collect started services

4. DEPLOY PHASE (Remote)
   ├── For each remote compose group:
   │   ├── Copy compose file to remote host
   │   ├── Detect compose command on remote
   │   ├── Execute: podman-compose -f <file> up -d
   │   └── Collect deployed services
   └── Handle failures with fallback

5. HEALTH CHECK PHASE
   ├── For each enabled service:
   │   ├── Execute health check (TCP/HTTP/gRPC)
   │   ├── Apply retry policy if configured
   │   └── Record result
   └── Aggregate results

6. SUMMARY PHASE
   ├── Count: started, failed, skipped, remote
```

```
   ├── Fail if required services are unhealthy
   └── Return BootSummary
```

Error Handling Strategy

```
┌──────────────────────────────────────────────────────────────────────────┐
│                          Error Handling
Strategy                                  │
└──────────────────────────────────────────────────────────────────────────┘
```

Error Type	Action	Recovery
Runtime unavailable	Warn, continue	Fall back to compose commands
Compose fail	Log error, continue	Mark service as failed
Health check fail	Retry with backoff	Mark unhealthy after max retries
SSH connection fail	Retry with backoff	Mark host offline, try next host
Scheduler fail	Log, use first host	Fall back to round-robin
Volume mount fail	Log warning	Continue without volume
Tunnel fail	Log, use direct	Fall back to direct connection

Performance Considerations

SSH Connection Pooling

```
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│                     ControlMaster Connection
Pool                                  │
└──────────────────────────────────────────────────────────────────────────┘
```

```
Without ControlMaster:
  Command 1: TCP connect + SSH handshake + exec (~500ms)
  Command 2: TCP connect + SSH handshake + exec (~500ms)
  Command 3: TCP connect + SSH handshake + exec (~500ms)
  Total: ~1500ms

With ControlMaster:
  Command 1: TCP connect + SSH handshake + exec (~500ms)
  Command 2: Reuse connection + exec (~50ms)
  Command 3: Reuse connection + exec (~50ms)
  Total: ~600ms (60% faster)
```

Resource Monitoring

Operation	Frequency	Overhead
Container list	10s	~5ms (local)
Container inspect	10s	~2ms per container
System metrics	10s	~1ms
Remote probe	60s	~50ms (SSH)

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